Questions-Gen — Three-Stage Training Architecture

Question data → Qwen3-14B SFT → GRPO reward → DeepSeek-R1 distillation → validation → publish

Primary flow
Fallback / feedback
Dashed = architecture zone
Box = module + responsibility

PROJECT OVERVIEW

Questions-Gen is a Qwen3-14B system for generating and evaluating mathematical competition problems through staged training.

The pipeline combines supervised fine-tuning, GRPO reward optimization, DeepSeek-R1 distillation, validation, and local deployment.

TRAINING DATA · documented mixture

- **Historical** — 50% — QuestionDataPreparer.py
- **Conditional Variants** — 30% — data_preparer.py
- **Innovative Types** — 20% — real_math_data_processor.py

Stage 1 · SFT

- **Qwen3-14B** — base checkpoint
- **QuestionsGenTrainer.py** — src/ + questions_gen/core/trainer.py
- **stage1-fp-merged** — Hugging Face

Stage 2 · GRPO

RewardCalculator.py

- **Difficulty** — 0.4 difficulty
- **Novelty** — 0.3
- **Rigor** — 0.2
- **Diversity** — 0.1

Optimization

- **GRPO Groups** — 8 problems/group
- **NoveltyConstraint.py** — duplication guard
- **stage2-merged-16bit** — variation generation

quality

Stage 3 · Distill

- **DeepSeekTeacher.py** — DeepSeek-R1
- **Teacher Signals** — difficulty · rigor · innovation · education
- **final-merged-16bit** — Hugging Face

VALIDATION · questions_gen/validation

- **model_validator.py** — single checkpoint
- **batch_validator.py** — algebra · calculus · geometry · number theory
- **quality_evaluator.py** — clarity · difficulty · originality

DEPLOYMENT · interfaces

- **main_cli.py** — validate · batch · quality
- **hf_utils.py** — three published stages
- **ollama_manager.py** — local inference
- **examples/** — quick_demo.py, demo_model_validation.py

Primary path: mixed problem set → SFT → GRPO reward and novelty → DeepSeek-R1 distillation → validate → Hugging Face / Ollama.